Mono3DR: Monocular Depth to Multi-view Reconstruction

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Abstract

Monocular depth estimators such as ZoeDepth produce strong per-image metric and relative depth predictions, but their outputs are often inconsistent across views, which limits their usefulness in multiview reconstruction. In this work, we study whether a lightweight alignment module can improve the multiview coherence of monocular depth without requiring ground-truth camera parameters at test time. We present DepthAlignNet, a lightweight multiview alignment model built on top of ZoeDepth monocular predictions. Our model takes paired RGB images and monocular depth predictions and jointly refines depth while estimating camera geometry. We train it with objectives that target pixel-wise reprojection consistency and geometric consistency. Experiments on ScanNet-derived sequences show that the proposed approach improves cross-view consistency over raw monocular predictions, while maintaining competitive depth quality compared to the median-scaled baseline. Although the aligned depths do not outperform median-scaled monocular depth on standard absolute metrics, they produce a more coherent multiview representation, which is the main goal of this work. The code is available under https://github.com/LaraTompa/mono-depth-to-3DR.git.

1. Introduction

Recent monocular depth estimators have made it possible to obtain detailed depth maps from a single RGB image [2, 3, 11, 13]. However, these methods are typically optimized per frame. As a result, depth predictions from neighboring views can be individually plausible yet geometrically inconsistent when projected into a common 3D frame. This is a major limitation for multi-view reconstruction, where depth estimates must agree across viewpoint changes.

The goal of this project is to improve the multiview consistency of monocular depth estimators using a lightweight network, with ZoeDepth as the main source of depth priors. Rather than trying to outperform strong single-view baselines on absolute depth accuracy, we focus on aligning predictions so that they better support reprojection, geometry, and photometric agreement across views. This makes the model more suitable as a practical component for 3D reconstruction pipelines.

Importantly, our goal is not to compete directly with recent foundation-style feedforward 3D reconstruction models such as VGGT, DUSt3R, or DepthAnythingV3 [7, 10, 12]. Instead, we study a more lightweight setting in which strong monocular depth priors are used as the starting point, and the network is designed to improve multiview consistency with minimal computational overhead. This makes the approach practical for scenarios where inference time and model size matter, while still aiming for competitive reconstruction-oriented behavior.

We explore several lightweight architectures during development, including depth-only and RGB-depth fusion variants, as well as outputing a depth map or a point map. The goal is to compare their ability to recover consistent depth across views. The final narrative of the paper should emphasize that the central contribution is not a new monocular depth backbone, but a consistency-aware refinement stage that makes existing monocular predictions more usable in multiview settings.

2. Related Work

2.1. Monocular Depth Estimation

There are two branches of work: Relative Monocular Depth Estimation (RMDE) and Metric Monocular Depth Estimation (MMDE). The first branch of work, relative depth estimation [9], deals with the large depth scale variation in multiple types of environments by factoring out the scale. As a result, disparity is sufficient for supervision; metric scale and camera parameters are not required and do not need to be consistent across datasets.

As for the second branch, the goal is to estimate depth in absolute physical units, i.e. meters [1, 14]. The advantage of predicting metric depth is the practical utility for many downstream applications in computer vision and robotics, such as mapping, planning, navigation, object recognition,

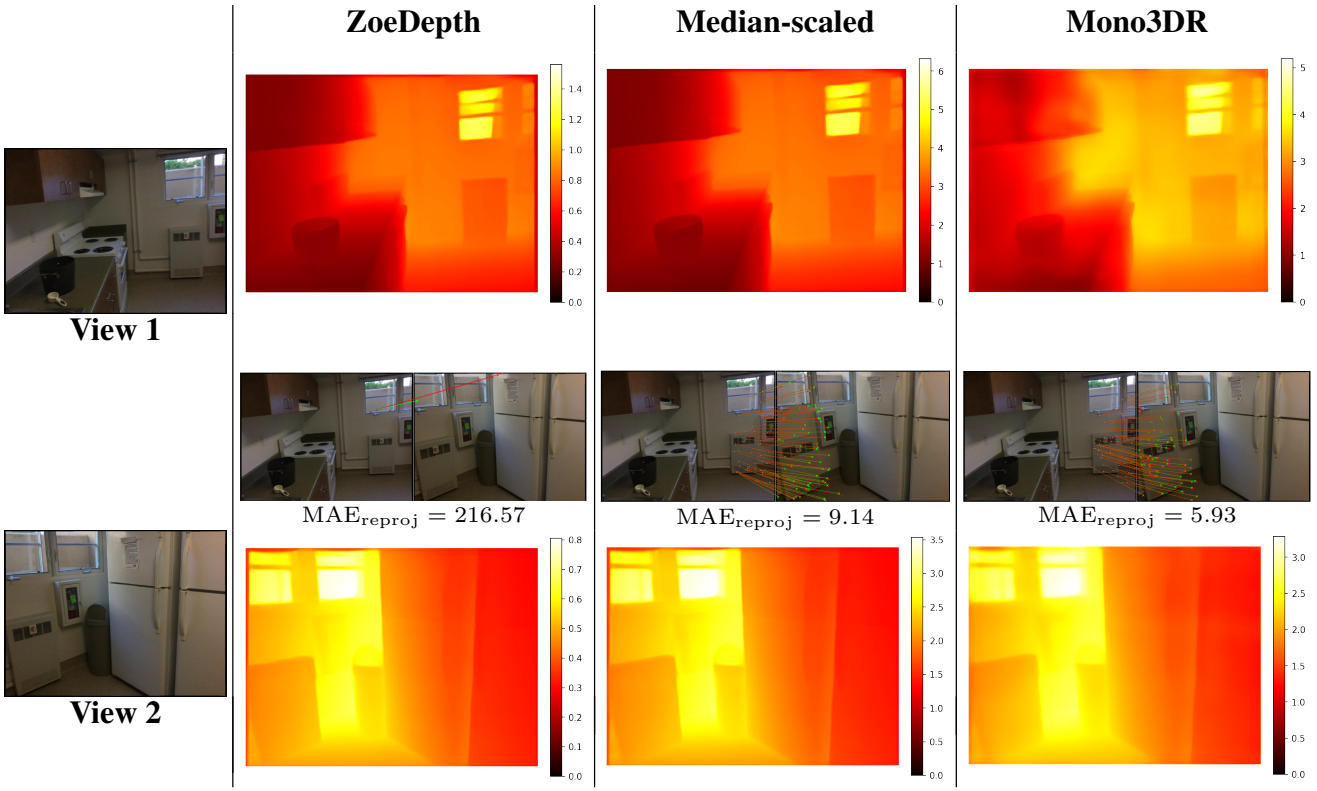


Figure 1. Qualitative comparison of raw ZoeDepth, median-scaled ZoeDepth, and Mono3DR for two views of the same scene. For each method, the predicted depth heatmaps of View 1 and View 2 are separated by the corresponding pixel-consistency visualization. The reprojection MAE is reported below each consistency visualization.

3D reconstruction, and image editing. However, training a single metric depth estimation model across multiple datasets often deteriorates the performance, especially when the collection includes images with large differences in depth scale, e.g. indoor and outdoor images. As a result, previous MMDE models usually overfit to specific datasets and do not generalize well to other datasets.

More recent approaches explore different techniques to overcome this. Metric3Dv2 [6] leverages surface normals as an auxilliary output to improve metric depth. DepthPro [3] is motivated in particular by novel view synthesis from as single image while focusing on boundary aaccuracy and jointly estimating the field of view from intermediate features of the depth prediction network.

In this work, we select ZoeDepth [2] as our fixed source of depth priors to train our network. This choice corresponds mostly to design, but is driven by its nature of combining both relative and metric approaches.

2.2. Feedforward 3D Reconstruction

DUSt3R [12] proposes a fundamental reformulation of multi-view 3D reconstruction. Classical Structure-from-Motion pipelines decompose the problem into discrete, error-prone stages: feature extraction, matching, relative pose estimation, triangulation, and bundle adjustment. DUSt3R collapses this entire pipeline into a single feed-forward network that, given an image pair directly regresses two dense pointmaps, where every pixel in each view is mapped to its corresponding 3D point expressed in the coordinate frame of view 1, which allows for emplicit encoding of the camera pose. While powerful, DUSt3R is computationally expensive ($O(N^2)$ pairs) and constitutes a bottleneck for large-scale scenes.

MASt3R [4] extends DUSt3R by augmenting the decoder with a dense local feature head that produces per-pixel, alongside the pointmaps. MASt3R is trained from a DUSt3R checkpoint, so the cross-attention decoder inherits a strong prior over inter-view geometric reasoning before the matching objective is introduced.

VGGT [10] addresses the scalability limitation of DUSt3R's pairwise paradigm by operating on all input frames simultaneously in a single forward pass. This amortises the cost of reconstruction to $O(N)$ and removes the need for any post-hoc global alignment. While VGGT represents a more scalable end-to-end solution, it requires all frames to be processed jointly and does not naturally incorporate external depth priors.

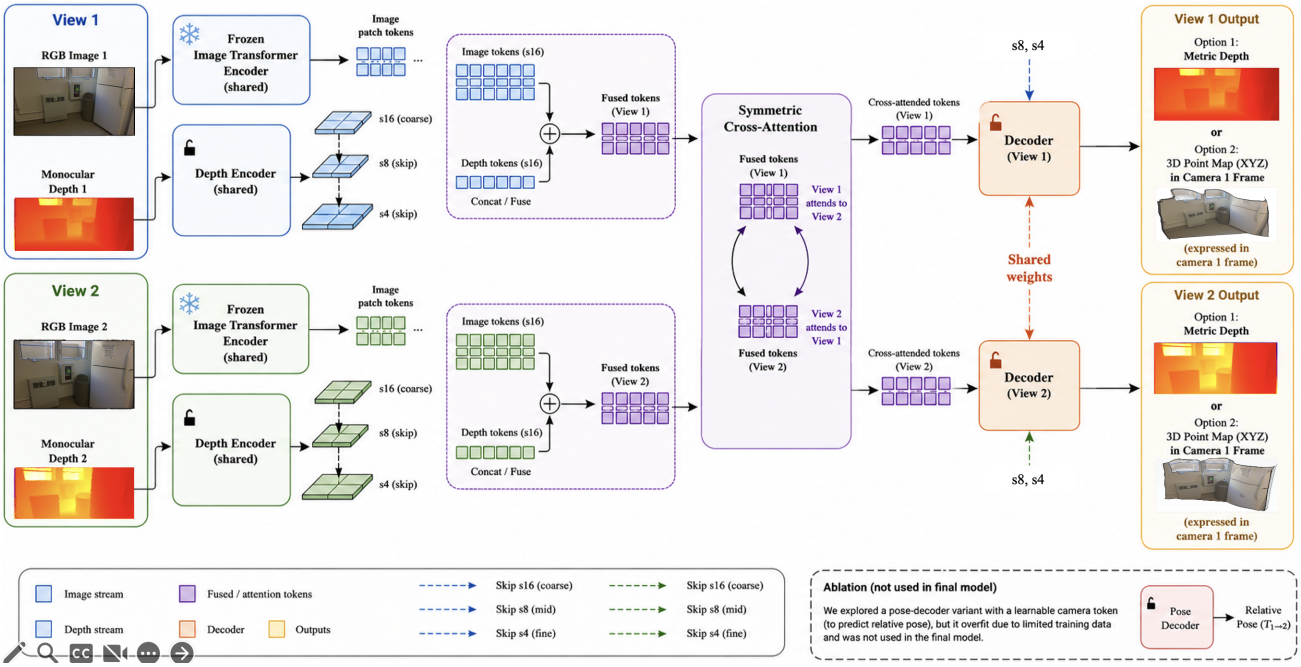


Figure 2. Overview of our two-view geometry network. Each view provides an RGB image and a monocular depth prediction. RGB images are processed by a **shared, frozen image transformer encoder** into patch tokens, while depth maps are processed by a **shared depth encoder** into multi-scale features (s16, s8, s4). At the coarse s16 level, image and depth tokens are fused per view before a **symmetric cross-attention** module lets the two views exchange information. A **shared decoder** combines the cross-attended tokens with skip connections from the depth encoder to regress dense geometry at full resolution, output as either a metric depth map or a 3D point map expressed in the coordinate frame of camera 1 (dashed placeholders indicate where output images can be inserted). A pose-decoder variant with a learnable camera token (bottom, dashed) was explored but discarded from the final model due to overfitting under limited training data. Image generated with ChatGPT.

3. Method

We design a shared two-stream model that processes RGB images and monocular depth predictions from two views, exchanges information through cross-attention, and outputs aligned depth or point maps plus camera parameters. The architecture is intentionally lightweight so that it can be used as a post-processing or adaptation module on top of existing monocular depth estimators. The key idea is to inject cross-view context at feature level, so the model can adjust each depth map in a way that better respects the other view's geometry. In addition, the pose and intrinsic predictions are learned jointly, which lets the model optimize multi-view consistency end-to-end rather than relying on fixed camera calibration.

3.1. Dataset and Sampling

To train and evaluate the Mono3DR, a custom temporal sampling strategy was implemented for the ScanNet RGB-D dataset[5]. In contrast to single-frame sampling, the proposed data loader generates short image sequences with controlled temporal spacing, allowing the model to learn from varying camera motions and view baselines.

The implementation operates directly on the ScanNet scene structure. During dataset initialization, valid scenes are identified based on the availability of RGB images, ground-truth depth maps, camera poses, and monocular depth estimates. Scenes are retained only if valid camera poses are available. Camera intrinsics are stored for each scene and reused during sampling.

For each training sample, a scene is selected and a sequence of a fixed number of frames is generated. The sequence is sampled using a center-anchored, strided strategy. A temporal stride is randomly selected from a predefined range, determining the distance between consecutive sampled frames in the original ScanNet sequence. For example, with a stride of 40–70 frames, consecutive observations are separated by a corresponding number of frames in the original approximately 30 Hz RGB-D sequence. This allows the training data to contain different camera motions and geometric baselines.

A center frame is then sampled such that the resulting sequence remains within the temporal boundaries of the scene. Frames are selected symmetrically around this center frame according to the sampled stride. If the scene is too short to accommodate the requested sequence and stride,

the implementation falls back to evenly spaced frame indices using linear interpolation. Consequently, every sample contains the required number of frames.

For each selected frame, the data loader provides the RGB image, the ScanNet sensor depth, the corresponding monocular depth estimate, and the camera pose. The monocular depth estimates are obtained from ZoeDepth and stored as PNG images in millimetres; they are converted to metres during loading. Invalid ScanNet poses, which may contain infinite values due to failed camera tracking, are replaced by identity transformations to prevent invalid numerical values from propagating through the training pipeline.

The sampling parameters, including the number of frames per sequence, number of samples per epoch, minimum and maximum temporal stride, and maximum number of scenes, are specified in the dataset configuration. This separates the sampling strategy from the implementation and allows different experimental settings to be evaluated without modifying the data loader.

3.2. Architecture

Our model takes a pair of (normalized) monocular depth predictions, together with the corresponding RGB images, and refines the scene geometry through a two-stream design as shown in detail in Figure 2. In each view, the depth input is processed by a shared depth convolutional encoder, using the ConvNeXt-architecture [8], that extracts a multi-scale feature pyramid, while the RGB image is processed by a frozen image transformer from a MASt3R checkpoint [4], that produces patch-level visaual tokens. At the coarsest spatial level, the image and depth representations are fused by aligning them on the same token grid and adding the image tokens to the depth tokens. The fused tokens from the two views are then passed through a cross-attention module in MASt3R style [12], which allows each view to aggregate geometric and appearance cues from the other view while preserving a shared latent representation across the pair.

Methods built on current two-view mod els inherit the asymmetric architecture with two separate decoders [4, 12], resulting in large model size. Importantly, the cross-attention module is chosen as symmetric, inspired on the work in [15]. Their results support this choice, which follows the main motivation of designing a lightweight network.

The cross-attended features are decoded together with skip connections from the depth encoder to recover a dense prediction at the target output resolution. Depending on the supervision and downstream task, the decoder can be configured to predict either a refined depth map or a dense 3D point map. In depth mode, the network outputs a single-channel metric depth map for each view. In point-map mode, it outputs a three-channel XYZ map in metric units. These 3D points are expressed in the coordinate frame of the first camera, so the first-view prediction is directly represented in camera-1 coordinates, while the second-view prediction is transformed into the same reference frame. This makes the output geometry directly comparable across the two views and convenient for multi-view consistency losses.

A pose-prediction variant was also explored, in which the cross-view decoder included a dedicated learnable camera token whose final embedding was passed to a small pose head to regress the relative transformation between the two views. Conceptually, this design follows the intuition that a global token can aggregate the scene-level correspondence information needed for pose estimation while sharing the same fused image-and-depth representation used for geometry prediction as explored in [12, 15]. In practice, however, this branch was not retained in the final model, as it showed clear signs of overfitting on our relatively small training set: training performance improved, but generalization to held-out scenes remained weak and unstable. We therefore attribute the limited success of the pose decoder primarily to data scarcity rather than to the overall performance of our architectural mechanism.

3.3. Losses

The total training objective is a weighted sum of the following depth-supervision and self-supervised consistency terms:

- **Scale-Invariant Depth Loss ($\mathcal{L}_{\text{depth}}$):** Adopted from Eigen-style depth supervision ($\lambda = 0.5$), measuring log-errors invariant to global scale:

$$\mathcal{L}_{\text{depth}} = \mathcal{L}_{\text{si_log}} = \frac{1}{N}\sum_i (\log \hat{d}_i - \log d_i)^2 - \frac{\lambda}{N^2}\left(\sum_i (\log \hat{d}_i - \log d_i)\right)^2 \tag{1}$$

- **MSE Depth Loss ($\mathcal{L}_{\text{depth_mse}}$):** Standard Euclidean metric applied strictly to valid depth regions ($d_i \in (10^{-3}, 80)$):

$$\mathcal{L}_{\text{depth_mse}} = \frac{1}{N}\sum_i \left(\hat{d}_i - d_i\right)^2 \tag{2}$$

- **Point-map loss ($\mathcal{L}_{\text{point}}$)**: For each pixel (i) with predicted point ($\hat{p}_i \in \mathbb{R}^3$) and ground-truth point ($p_i \in \mathbb{R}^3$), the per-pixel error is the mean absolute distance over the (x,y,z) channels:

$$\ell_i = \frac{1}{3}\sum_{c\in x,y,z} \left|\hat{p}_i^{(c)} - p_i^{(c)}\right|. \tag{3}$$

With predicted confidence ($c_i \in (0, 1]$) and uncertainty s_i clip $(-\log c_i, s_{\min}, s_{\max})$, each term is weighted as

$$\mathcal{L}_{\text{point}} \frac{1}{|V|}\sum_{i\in V} \left(e^{-s_i}\ell_i + s_i\right), \tag{4}$$

where (V) is the set of valid pixels (finite, in-range ground-truth depth).

- **Pixel Consistency Loss ($\mathcal{L}_{\text{pix}}$):** A bidirectional multiview reprojection term ($\mathcal{L}_{\text{pix}} = \frac{1}{2}(\mathcal{L}_{12} + \mathcal{L}_{21})$). For direction $1 \to 2$:

$$\mathcal{L}_{12} = \frac{1}{N_{\text{val}}} \sum_{p} \left\| \pi \left(K, T_{12} \cdot \Pi(p, \hat{d}) \right) - \pi \left(K, T_{12} \cdot \Pi(p, d_{\text{gt}}) \right) \right\|^2 \tag{5}$$

  computed using normalized pixel coordinates and a capped distance.

- **Geometric Consistency Loss ($\mathcal{L}_{\text{geo}}$):** A ViSTA-style 3D warp consistency term ($\mathcal{L}_{\text{geo}} = \frac{1}{2}(\mathcal{L}_{12}^{\text{geo}} + \mathcal{L}_{21}^{\text{geo}})$):

$$\mathcal{L}_{12}^{\text{geo}} = \frac{1}{N_{\text{val}}} \sum_{x} \left\| Q_{ij}(x) - \hat{P}_j(C_{ij}(x)) \right\|^2 \tag{6}$$

  where $Q_{ij}(x)$ is the GT 3D point of pixel x in frame j after transformation, and $\hat{P}_j(C_{ij}(x))$ is the 3D point obtained via bilinear sampling of predicted depth at the warped pixel.

Other losses used for training the pose head included rotation, translation, and a round-trip (inverse) consistency check. The rotation term measures the geodesic angular error between the predicted and ground-truth relative rotation, while the translation term compares the predicted translation direction with the ground-truth direction, with the scale handled separately to avoid instability from small baselines. An additional identity loss enforces that the forward and backward pose estimates are consistent with each other, so the network does not learn mutually contradictory transforms. In the implementation, uncertainty-aware weighting is used for the pose and intrinsics terms during the main training phase, but this is disabled during the initial warmup period so the network first stabilizes its pose geometry before learning confidence estimates.

4. Experiments

4.1. Experimental Setup

We evaluate on ScanNet-derived sampled sequences and compare aligned predictions against median-scaled monocular depths. Standard depth metrics show that the baseline remains slightly better on absolute accuracy, but the proposed method achieves comparable consistency across views with a strictly limited train dataset size, thus suggesting that scalability might reduce the breach between the performance with respect to the baseline and potentially achieve superior results.

4.2. Results

On the test set, aligned depths achieve around 0.134–0.136 AbsRel and 0.267–0.273 RMSE, while the median-scaled monocular baseline reaches around 0.070 AbsRel and 0.194 RMSE. In contrast, consistency improves notably: pixel-consistency MAE drops from about 22.1 to 37.8 in the mono-scaled baseline comparison space, geometric inconsistency improves from about 0.215 m to 0.287 m depending on the metric definition, and photometric SSIM/L2 also move in the right direction. The important message is that the model sacrifices some single-view accuracy to gain a more coherent multiview depth field.

4.3. Implementation Details

All models are trained using the AdamW optimizer with a learning rate of (1×10^{-5}) and a weight decay of (1×10^{-4}). The learning rate follows a cosine decay schedule. Training is performed with a batch size of 4 and gradient accumulation over 2 steps, resulting in an effective batch size of 8. Gradients are clipped to a maximum norm of 1.0. Training is conducted for up to 60 epochs, with early stopping based on the validation loss.

To account for the heterogeneous depth distributions across different scenes, depth and 3D point-map targets are normalized using median and scale statistics estimated from the training split. The same normalization parameters are then applied to the validation and test splits, ensuring that information from the evaluation data does not influence the training procedure.

The training objective consists of the losses described in Section 3. The primary supervision is introduced first, followed by the additional consistency terms after a warm-up period of five epochs.

For evaluation, standard monocular depth estimation metrics are computed over valid ground-truth pixels, using the same validity mask as during training. The reported metrics include Absolute Relative Error (AbsRel), Root Mean Squared Error (RMSE), and threshold accuracies δ_1, δ_2, and δ_3.

As a reference, the raw monocular depth predictions are also evaluated after median scaling. For each view, the prediction is rescaled using the median ratio between the ground-truth and predicted depths before computing the same evaluation metrics.

5. Conclusion

We investigated a multi-view-consistent approach to monocular depth alignment. Although the proposed model does not yet outperform the median-scaled monocular baseline in absolute depth accuracy, the results provide encouraging evidence for the underlying approach.

The remaining gap in AbsRel and RMSE should be interpreted in the context of the computational constraints of the experiments. Training was limited to a relatively small subset of ScanNet scenes and a modest number of epochs,

Table 1. **Quantitative Evaluation Results.** Comparison of depth accuracy and multi-value consistency across methods. Metrics represent average performance across 3315 evaluated frame pairs.

	Depth Quality Metrics		Multi-View Consistency Metrics		
Method	**RMSE ↓**	**Abs Rel ↓**	**PC RMSE (px) ↓**	**Geo L2 (m) ↓**	**SSIM ↑**
Monocular Baseline	1.4235	0.7112	193.02	1.8764	0.4635
Monocular Baseline (Scaled)	0.1938	0.0708	31.88	0.2151	0.6391
Mono3DR v1 (depth + image)	0.2698	0.1352	44.95	0.2866	0.6249
Mono3DR v2 (pointmap + image)	0.2813	0.1299	58.83	0.3440	0.5894
Mono3DR v3 (depth)	0.3098	0.1575	56.17	0.3356	0.5894
Mono3DR v4 (pointmap)	0.3259	0.1616	61.15	0.3839	0.5925

restricting both the diversity of the training data and the degree of model convergence. The temporal sampling strategy was similarly influenced by these constraints, using relatively large temporal strides to keep data-loading and training costs manageable rather than maximizing view coverage.

Despite these limitations, the proposed model achieves multi-view consistency comparable to the baseline, suggesting that the geometric constraints can be added without compromising cross-view consistency. Future work should therefore focus on increasing the training data and compute budget, extending the training schedule, and investigating denser or overlap-aware sampling strategies. These extensions would provide a more comprehensive assessment of the method's generalization capabilities and its potential to close or overcome the observed accuracy gap.

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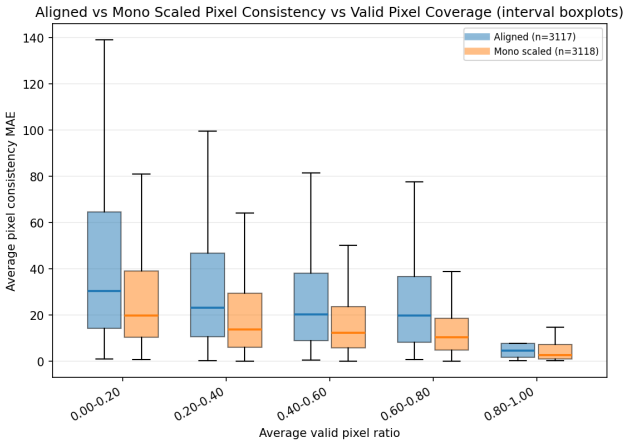
Appendix

To better understand the relationship between multiview consistency errors and other geometric factors on the test set, we generated scatter plots (with linear trend lines) and corresponding interval boxplots for several metric pairs, comparing our aligned model output against the median-scaled monocular baseline. Each point corresponds to one evaluated frame pair aggregated over all test scenes (around 3300 ablation pairs).

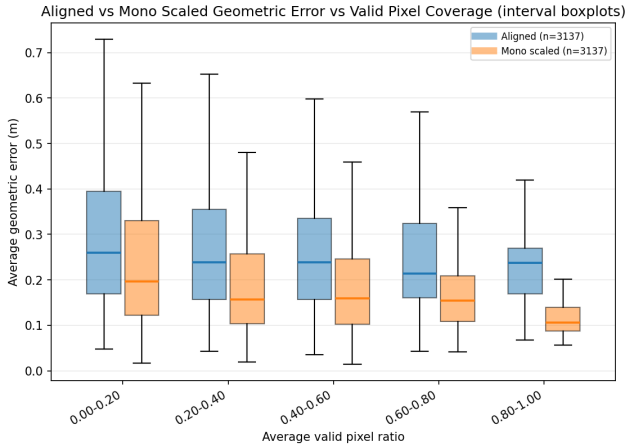
Four relationships were analyzed:
Pixel-consistency MAE vs. valid pixel ratio tests whether reprojection error is driven by limited overlap/occlusion between views rather than by depth quality itself.
Geometric error vs. valid pixel ratio which is the 3-D counterpart of the above, measuring whether metric-space consistency also degrades with reduced view overlap.
Pixel-consistency MAE vs. geometric error checks whether the 2-D reprojection metric and the 3-D warp-consistency metric agree on which pairs are inconsistent.
Photometric L2 vs. valid pixel ratio relates appearance-based consistency (warped image difference) to view overlap.
For each relationship we report the Pearson correlation coefficient for both the aligned model and the mono-scaled baseline.

Key Findings

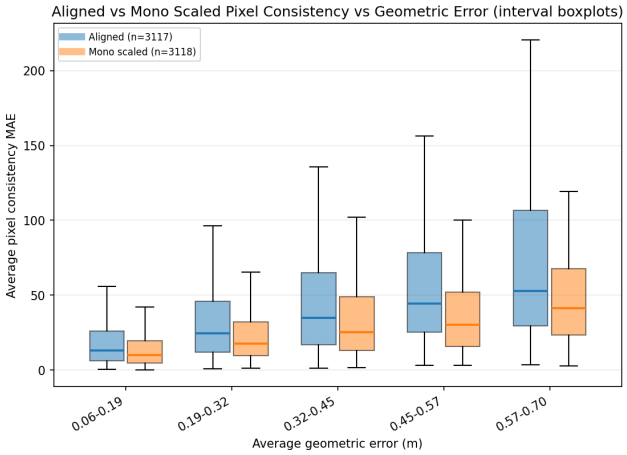
Valid pixel ratio is weakly-to-moderately negatively correlated with all three consistency errors, indicating that pairs with less shared visible area tend to have somewhat higher reprojection, geometric, and photometric error, as expected, but the effect is not dominant, most of the error variance is not explained by overlap alone. Notably, the mono-scaled baseline is consistently more sensitive to valid-ratio changes than our aligned model, suggesting our model's consistency is comparatively more robust to reduced overlap. Pixel-consistency MAE and geometric error are moderately positively correlated and nearly identical between the aligned and baseline models, confirming that the two consistency metrics capture a related but not fully overlapping notion of multiview agreement, and that this relationship is largely independent of which depth source is used. Overall, the analysis supports treating pixel-consistency, geometric, and photometric errors as complementary rather than redundant diagnostics, and indicates that our aligned model degrades more gracefully than the monocular baseline as view overlap decreases.



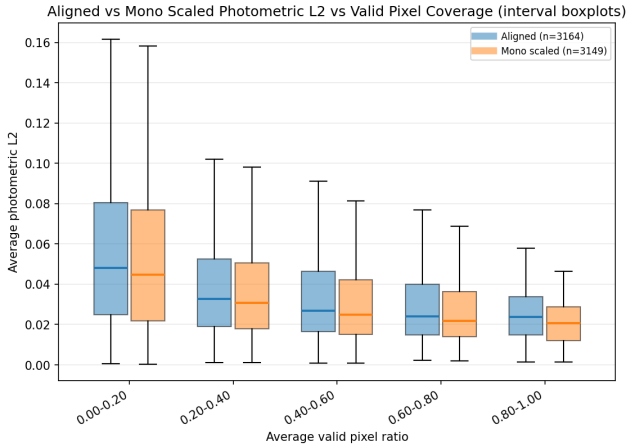
(a) Pixel-consistency MAE vs. valid pixel ratio.



(b) Geometric error vs. valid pixel ratio.



(c) Pixel-consistency MAE vs. geometric error.



(d) Photometric L2 vs. valid pixel ratio.

Figure 3. Correlation between multiview consistency metrics and view-overlap/geometric factors on the test set, comparing our aligned model against the median-scaled monocular baseline (Pearson r reported per series; trend lines from a linear fit). See the appendix for a full discussion.